We are delighted to welcome you all to the

Free Summer Course on AI & Ethical Hacking

organized by

SRI GANESH COLLEGE OF ENGINEERING & TECHNOLOGY

and

PSV COLLEGE OF ARTS & SCIENCE

DAY 1

WEB DEVELOPMENT

AGENDA

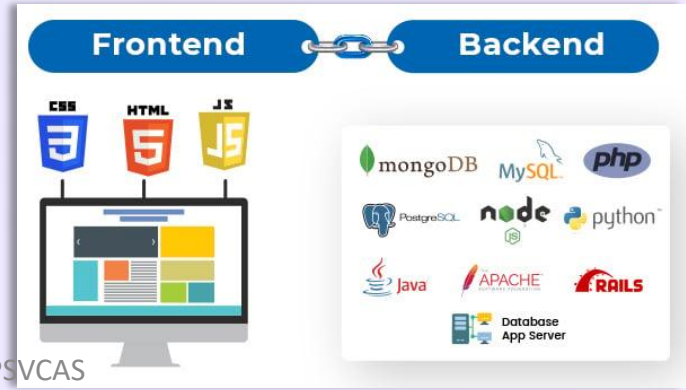
- What is Web Development?
- Types of Web Development
- Key Components of a Website
- How Websites Work? (Client-Server Model)
- Popular Web Technologies
- Static vs Dynamic Websites
- Career Opportunities in Web Development
- HTML and it’s tags
- CSS

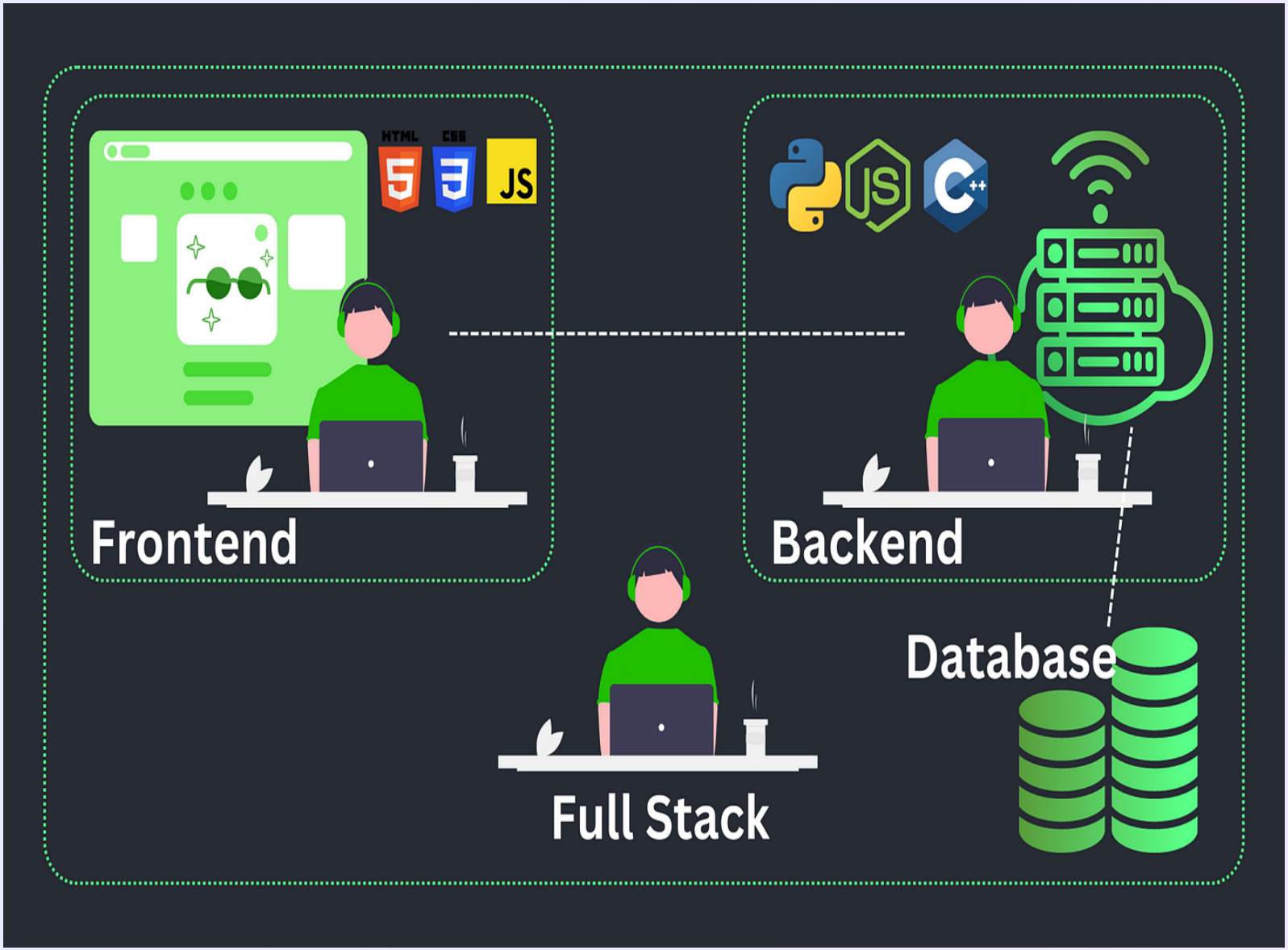
What is Web Development?

- Web development is the process of creating websites and web applications.
- It involves building, maintaining, and managing websites.
- It includes web design, web content development, and coding.

Types of Web Development

- **Frontend Development:** What users see and interact with (HTML, CSS, JavaScript).
- **Backend Development:** Server-side logic, databases, APIs (PHP, Node.js, Python, etc.).
- **Full Stack Development:** Combines both frontend and backend.




HTML
CSS
JS
Frontend
Backend
Database
Full Stack

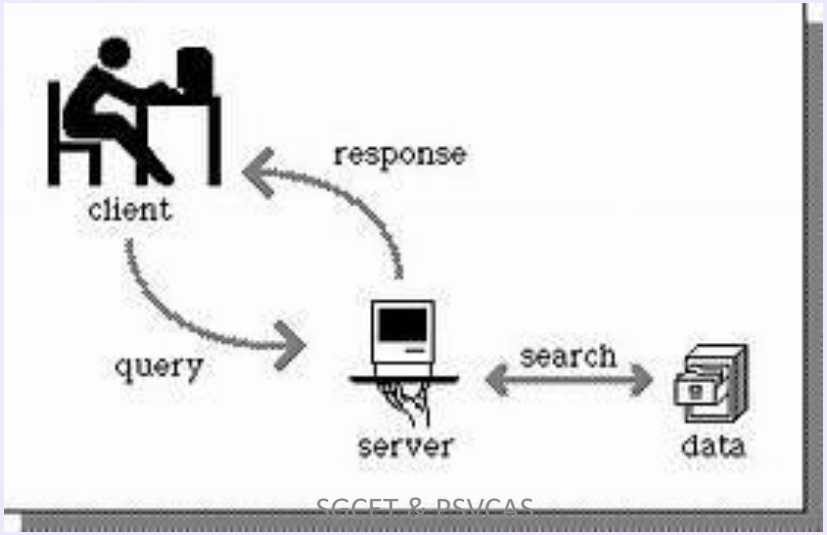
Key Components of a Website

- **HTML** (Hyper Text Markup Language) – Structure
- **CSS** (Cascading Style Sheets) – Styling
- **JavaScript** – Interactivity
- **Web Server** – Hosts the website
- **Database** – Stores data (MySQL, MongoDB)

How Websites Work?

(Client-Server Model)

- User sends a request through a browser (client).
- Server receives the request and sends back data.
- Browser renders and displays the webpage.



Popular Web Technologies

- **Frontend**: HTML, CSS, JavaScript, React, Angular
- **Backend:** Node.js, Django, Ruby on Rails
- **Databases:** MySQL, MongoDB, PostgreSQL
- **Tools:** Git, VS Code, Browser DevTools

Frontend

CSS
JavaScript
React
Angular
Git

Backend

node JS
django
React

Databases

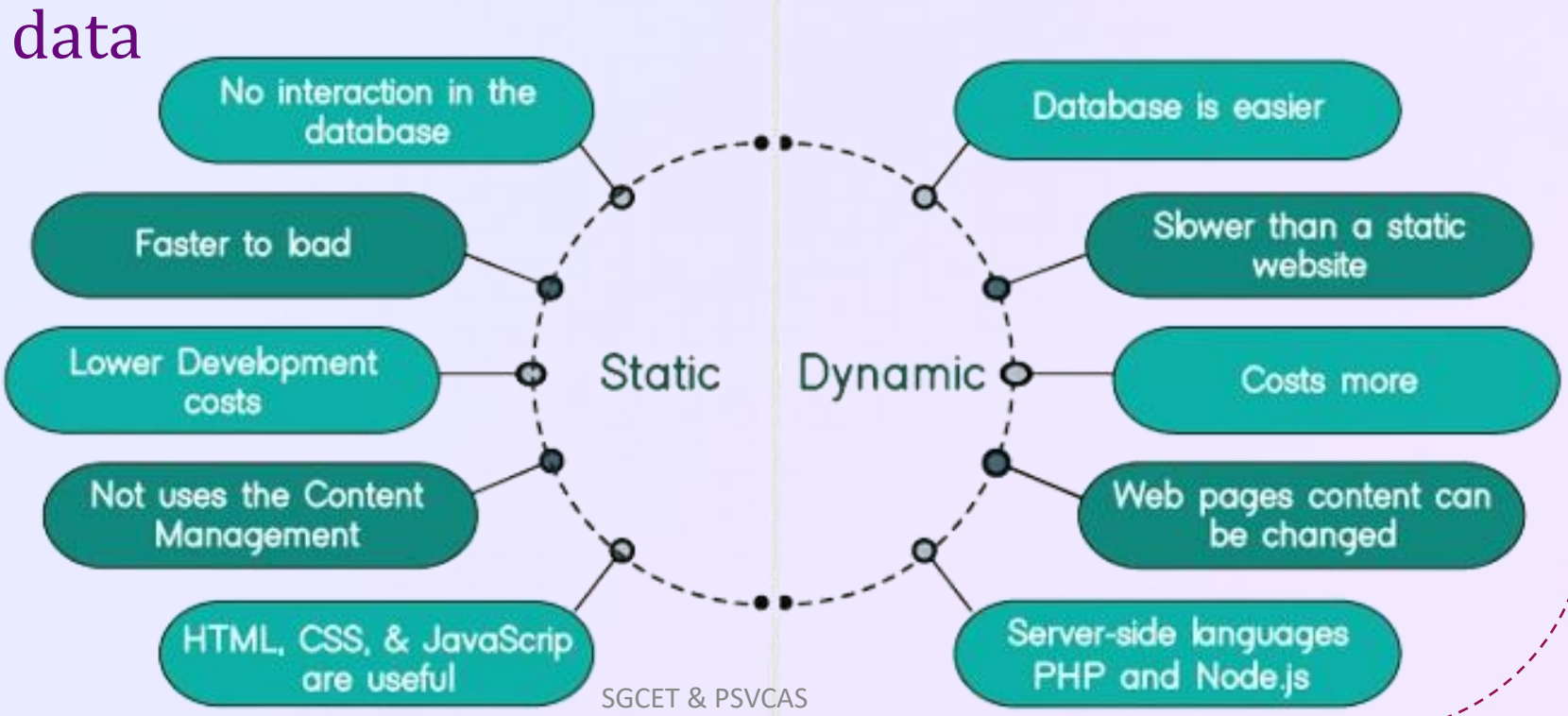
mongoDB
MySQL

Tools

VS Code
Browser DevTools

Static vs Dynamic Websites

- **Static:** Simple pages, same content for every user
- **Dynamic:** Content changes based on user interaction or data



Career Opportunities in Web Development

- Frontend Developer
- Backend Developer
- Full Stack Developer
- UI/UX Designer
- Web Tester / QA Engineer

HTML and it's tags

What is HTML?

- Stands for **Hyper Text Markup Language.**
- The standard language for creating web pages.
- Defines the structure of web content.

Note:
HTML is not a programming language;
it's a markup language.

Key HTML Elements:

- <!DOCTYPE html> – Declares the document type
- <html> – Root element
- <head> – Metadata, title, styles
- <body> – Visible page content

Common Tags:

- <h1> to <h6> – Headings
- <p> – Paragraph
- <a href=""> – Hyperlink
- <img src=""> – Image

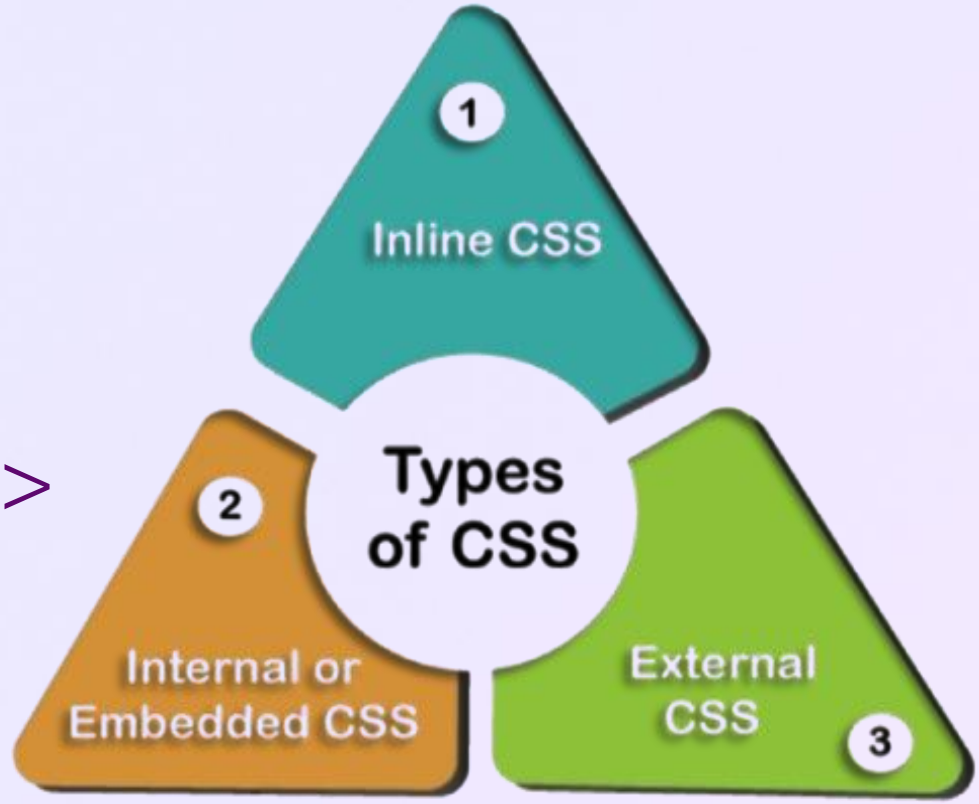
CSS

What is CSS?

- Stands for **Cascading Style Sheets.**
- Used to style and layout web pages.
- Works with HTML to enhance visual appearance.

Types of CSS

- **Inline:** Inside HTML elements using the style attribute
- **Internal:** Within <style> tags inside the <head>
- **External:** Linked using <link rel="stylesheet">



Common Properties:

- color, background-color
- font-size, font-family
- margin, padding, border
- width, height, display

Basic syntax:

```
selector {
  property: value;
}
```

Example:

```
p {
  color: blue;
  font-size: 16px;
}
```

THANK YOU